

EN.553.285 Python Setup Guide

Miniconda + Python + Jupyter

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Goal: By the end of this guide, you will have:

- Miniconda installed
- A class environment created (with Python)
- Jupyter Notebook installed inside that environment
- NumPy, SciPy, Matplotlib, and Pandas installed and working

Important rule: For this course, always **activate the class environment first**, then run Jupyter.

1 Step 0: Choose Your Operating System

This guide has separate steps for:

- **macOS** (Apple Silicon M1/M2/M3 or Intel)
- **Windows**

If you are unsure which Mac you have:

- Click **Apple** menu → **About This Mac**
- If it says **Chip:** **Apple M...** you have Apple Silicon (arm64)
- If it says **Processor:** **Intel...** you have Intel (x86_64)

2 macOS Installation (Recommended)

2.1 1. Download Miniconda (macOS)

Download Miniconda from:

<https://docs.conda.io/en/latest/miniconda.html>

Choose the installer that matches your machine:

- **Apple Silicon (M1/M2/M3):** Miniconda3 macOS arm64
- **Intel Mac:** Miniconda3 macOS x86_64

You will download a file that ends in `.sh` (for example: `Miniconda3-latest-MacOSX-arm64.sh`).

2.2 2. Install Miniconda (macOS)

Open the **Terminal** app and run:

```
cd ~/Downloads  
ls
```

Find the Miniconda file name in the list, then run (replace the filename if needed):

```
bash Miniconda3-latest-MacOSX-arm64.sh
```

During installation:

- Accept the license
- Accept the default install location
- When asked to initialize conda, choose **yes**

2.3 3. Restart Terminal and verify

Close Terminal completely and reopen it.

Check that conda is installed:

```
conda --version
```

If you see a version number, Miniconda is working.

3 Windows Installation

3.1 1. Download Miniconda (Windows)

Download Miniconda from:

<https://docs.conda.io/en/latest/miniconda.html>

Choose:

- **Windows 64-bit (most common):** Miniconda3 Windows x86_64

You will download a file that ends in **.exe**.

3.2 2. Install Miniconda (Windows)

Run the installer and follow prompts.

Recommended choices:

- Install for “Just Me” (fine for most students)
- Default install location is fine
- If asked about adding conda to PATH: leaving default is usually fine.

3.3 3. Open “Anaconda Prompt” and verify

After installation, open:

Anaconda Prompt (Miniconda3)

Verify conda:

```
conda --version
```

4 Create the Class Environment (macOS or Windows)

From this point onward, the commands are the same on both systems:

- macOS: use **Terminal**
- Windows: use **Anaconda Prompt (Miniconda3)**

4.1 1. Configure conda-forge (recommended)

This helps install up-to-date scientific packages reliably:

```
conda config --add channels conda-forge
conda config --set channel_priority strict
```

4.2 2. Create the environment

Create an environment called `sci-python` with Python 3.12:

```
conda create -n sci-python python=3.12 -y
```

4.3 3. Activate the environment

```
conda activate sci-python
```

You should now see `(sci-python)` at the start of your terminal prompt.

4.4 4. Install course packages + Jupyter

Install Jupyter and the core libraries we will use:

```
conda install jupyter numpy scipy matplotlib pandas -y
```

5 Launch Jupyter Notebook in Your Browser

5.1 1. Start Jupyter

Make sure `(sci-python)` is active, then run:

```
jupyter notebook
```

This should open a browser window with the Jupyter file browser.

5.2 2. If it does not open automatically

Look in your terminal for a URL like:

```
http://localhost:8888/?token=...
```

Copy and paste it into your browser.

5.3 3. Stop Jupyter

To stop the server, go back to the terminal and press:

Ctrl + C

Then type y if it asks for confirmation.

6 Sanity Check (Very Important)

6.1 1. Confirm Jupyter is using the correct Python

In Jupyter:

- Click New → Python 3
- Run the following cell:

```
import sys
print(sys.executable)
```

You should see a path that contains something like:

- .../envs/sci-python/... (macOS)
- ...
envs
sci-python
... (Windows)

6.2 2. Confirm packages import

Run:

```
import numpy as np
import scipy
import matplotlib
import pandas as pd

print("NumPy:", np.__version__)
print("SciPy:", scipy.__version__)
print("Matplotlib:", matplotlib.__version__)
print("Pandas:", pd.__version__)
```

If this runs with no errors, your environment is ready.

7 Everyday Workflow for This Class

Every time you want to work:

```
conda activate sci-python
jupyter notebook
```

Do not install packages without activating the environment first.

8 Common Problems and Fixes

8.1 Problem: conda: command not found (macOS)

Fix:

- Close Terminal and reopen it
- If still broken, reinstall Miniconda and answer **yes** to “initialize conda”

8.2 Problem: Jupyter opens but imports fail

Almost always: you launched Jupyter from the wrong environment.

Fix:

```
conda activate sci-python
jupyter notebook
```

Then re-check:

```
import sys
print(sys.executable)
```

8.3 Problem: Port already in use

If you see an error about port 8888, run:

```
jupyter notebook --port 8890
```

9 Optional: Installing VS Code (Recommended)

VS Code makes it easier to edit `.py` files and organize projects.

Download:

<https://code.visualstudio.com/>

Once installed:

- Install the **Python** extension in VS Code
- Select the `sci-python` interpreter when prompted